

Public Cloud Services Container Services



The Container Iceberg

<https://dev.to/pulumi/the-cloud-container-iceberg-at-least-105-ways-to-run-a-container-in-the-cloud-3kb5>



Compute, Product Classes in AWS

Virtual Machines (VMs)	Containers	Platform as a Service (PaaS)	Serverless	Specialized Solutions
 Amazon Elastic Compute Cloud (Amazon EC2)	 Amazon Elastic Container Service (Amazon ECS)	 AWS Elastic Beanstalk	 AWS Lambda	 AWS Outposts
 Amazon Lightsail			 AWS Fargate	 AWS Batch

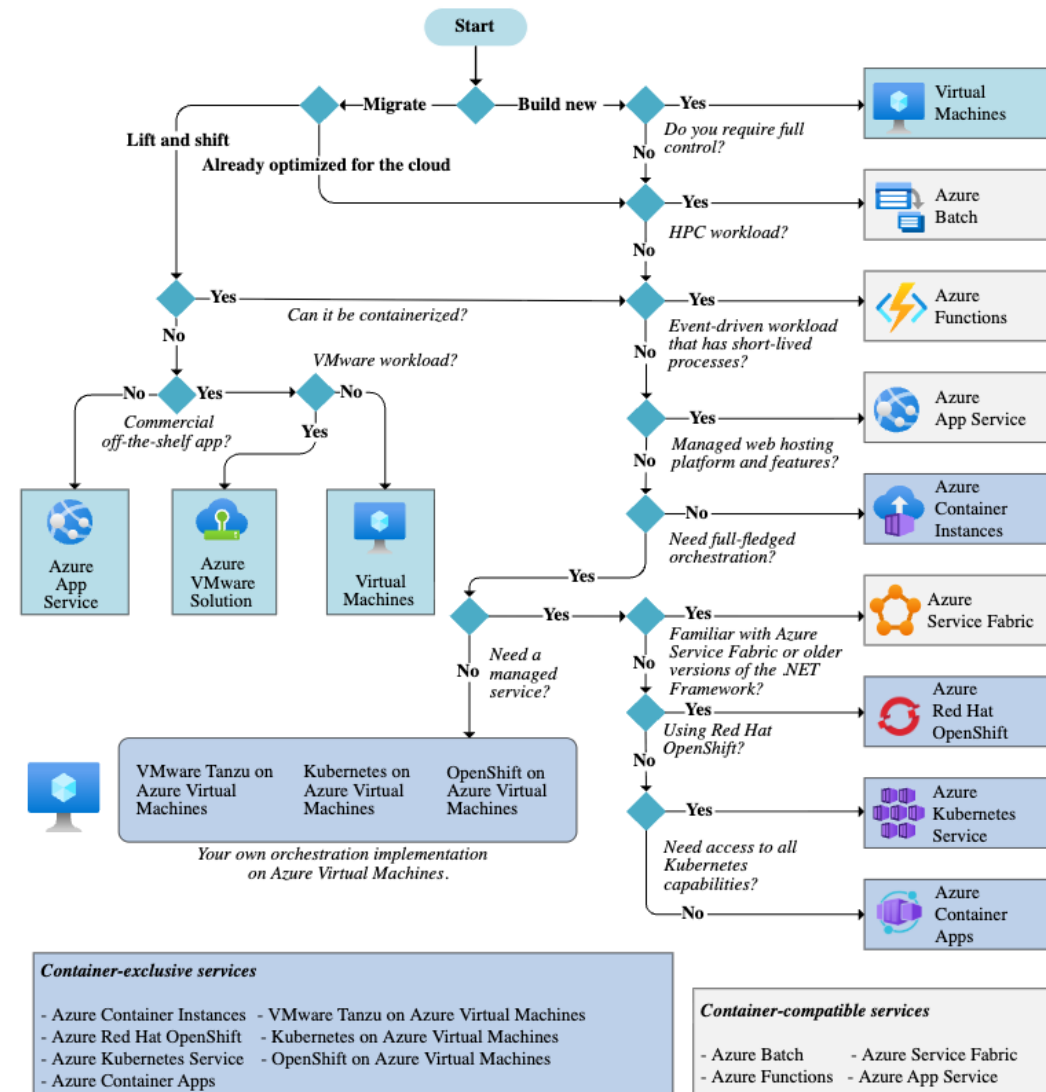
Higher infrastructure control and customization

Faster application deployment

Fully managed services

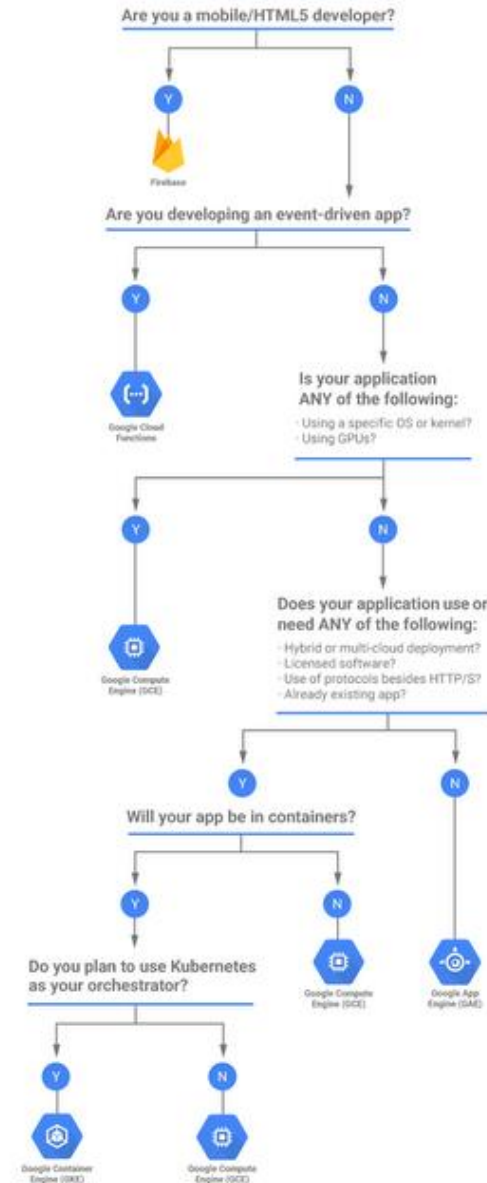
Compute, Product Classes in Azure

<https://learn.microsoft.com/en-us/azure/architecture/guide/technology-choices/compute-decision-tree>



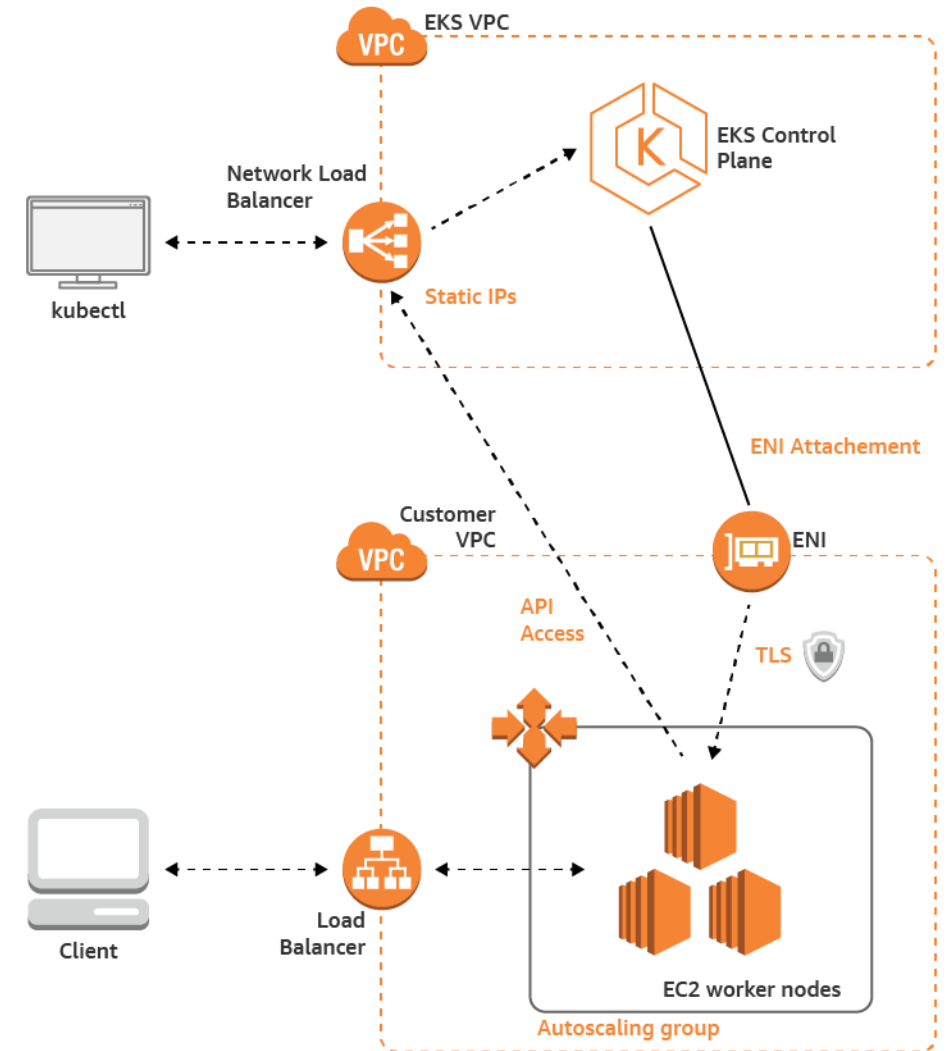
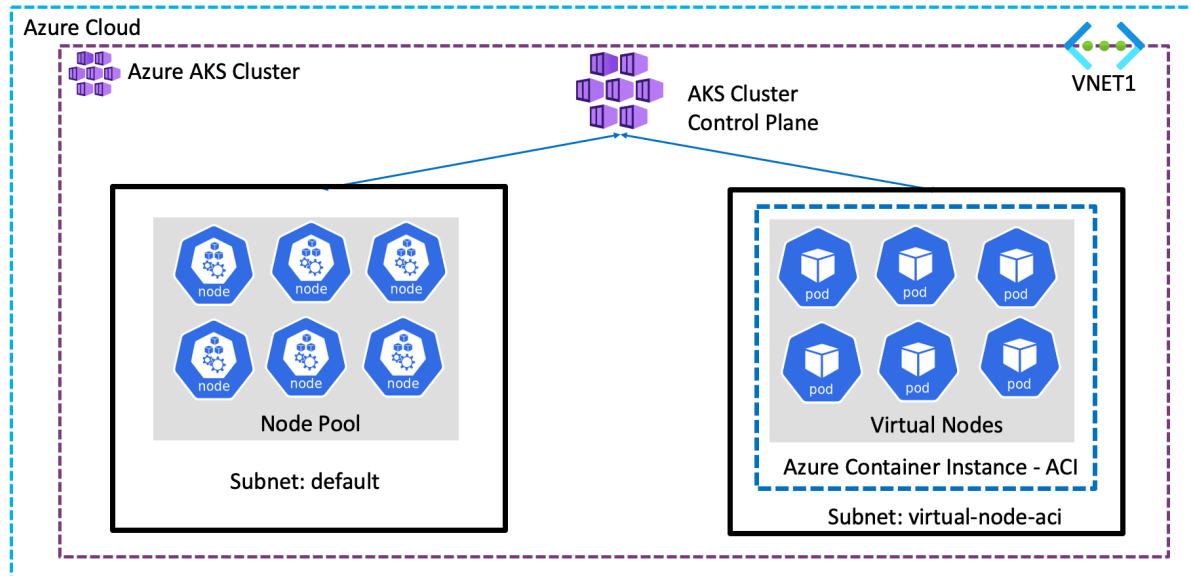
Compute, Product Classes in Google Cloud

<https://cloud.google.com/blog/products/compute/choosing-the-right-compute-option-in-gcp-a-decision-tree?hl=en>



Managed Kubernetes,

Azure AKS - Virtual Nodes

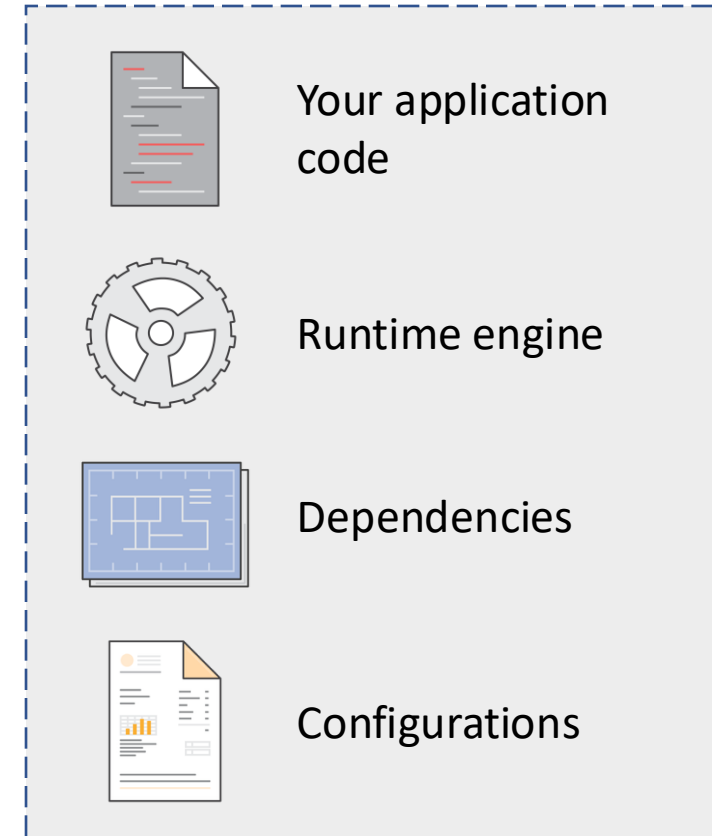


Running Container Natively

Container as atomic unit for deployment:

- Container all Information / Software for Software to run
- ..including Dependencies
- ..including Runtimes

What do you need from an infrastructure perspective to run a container?



Compute Engines, e.g. ECS

Elastic Container Service:

- Launching 2014
- Removed need to provision IaaS by yourself (and to maintain it)
- Handles multiple IaaS-instances from a container perspective (even though Reserved / Spot Instances are possible)
- Can work with “serverless” instances (Fargate).



Amazon Elastic
Container Service
(Amazon ECS)

Definition of an ECS instance

Container Definition:

- bases by the image itself

Task Definition:

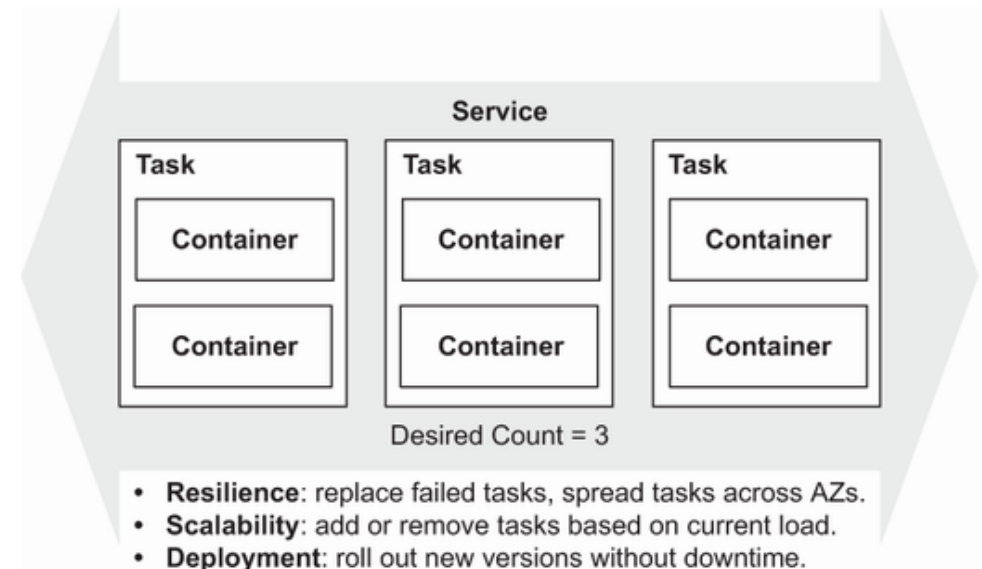
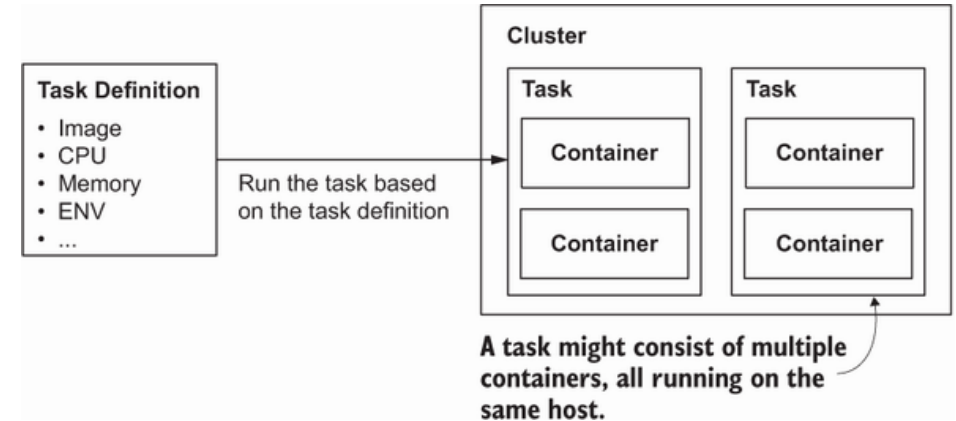
- Relies on one or multiple containers
- Revisioned regarding settings, etc.

Service:

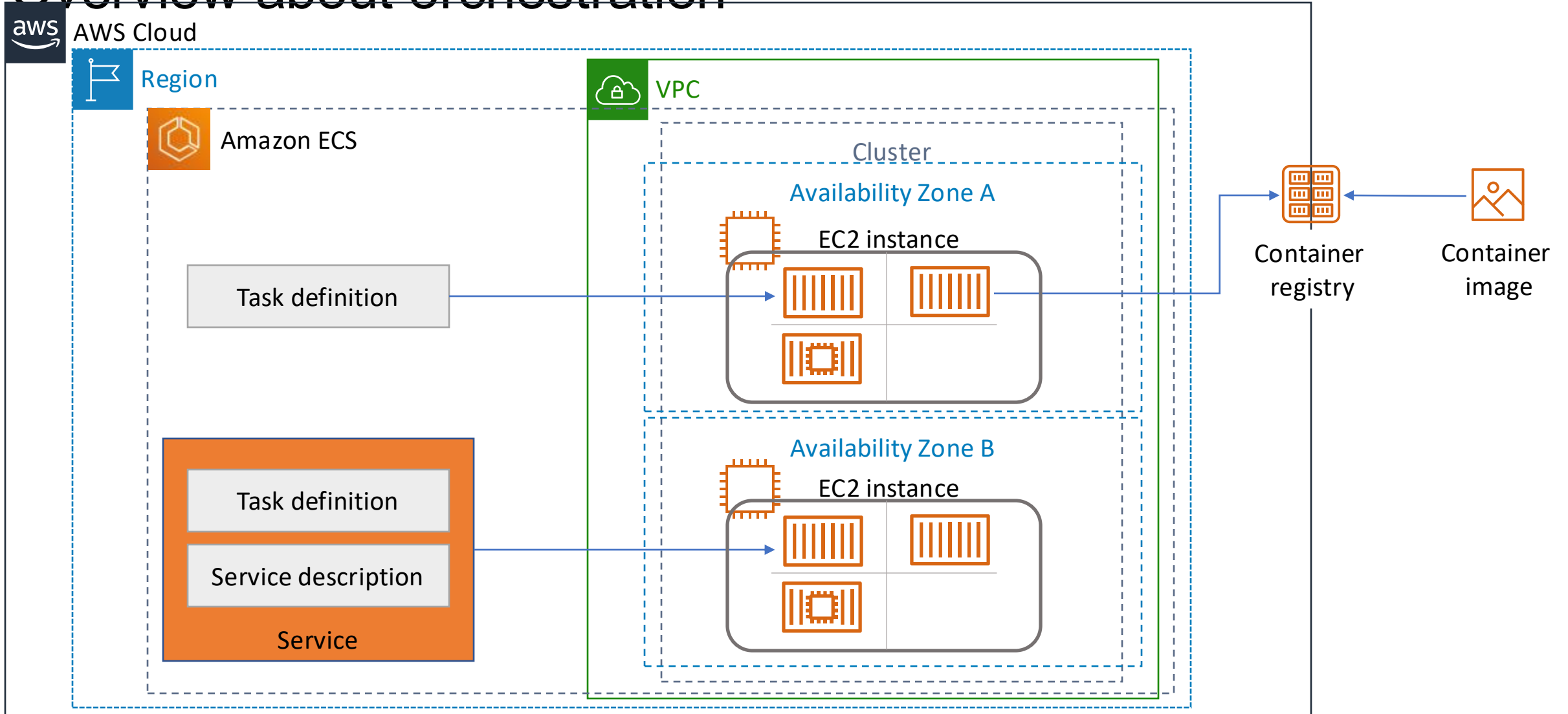
- Defining count, auto scaling, access for Task

Cluster:

- Summarizing multiple container services



Overview about orchestration



Serverless VS IaaS

